Experiment:14 Numerical Simulation and Spectral Analysis of a Microwave Signal for Frequency-Domain Characterization Using Spectrum-Analyzer Concepts

Aim: To simulate a microwave signal and analyze its frequency spectrum using MATLAB based on spectrum-analyzer concepts.

Apparatus:

Computer/Laptop

Matlab Software

Procedure:

- Open MATLAB and create a new script.
- Define the sampling frequency, simulation time, and microwave signal frequencies.
- Generate a composite microwave signal containing 1 GHz, 2 GHz, and 3 GHz components.
- Add random noise to the generated signal.
- Plot the signal in the time domain.
- Apply the Fast Fourier Transform (FFT) to convert the signal into the frequency domain.
- Plot the frequency spectrum and identify the dominant frequency components.
- Run the program and observe the time-domain waveform, spectrum, and detected peak frequencies.

Code:

```
clc;

clear;

close all;

% Signal parameters

Fs = 10e9;

T = 2e-6;

t = 0:1/Fs:T-1/Fs;

f1 = 1e9;

f2 = 2e9;

f3 = 3e9;

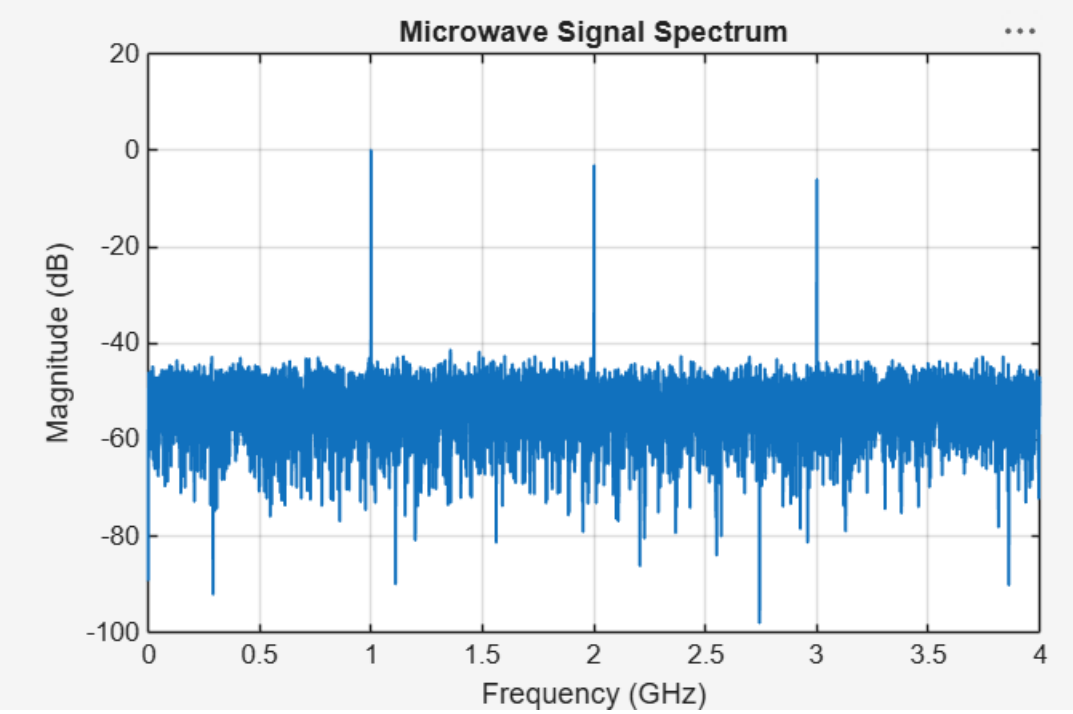
% Generate microwave signal
```

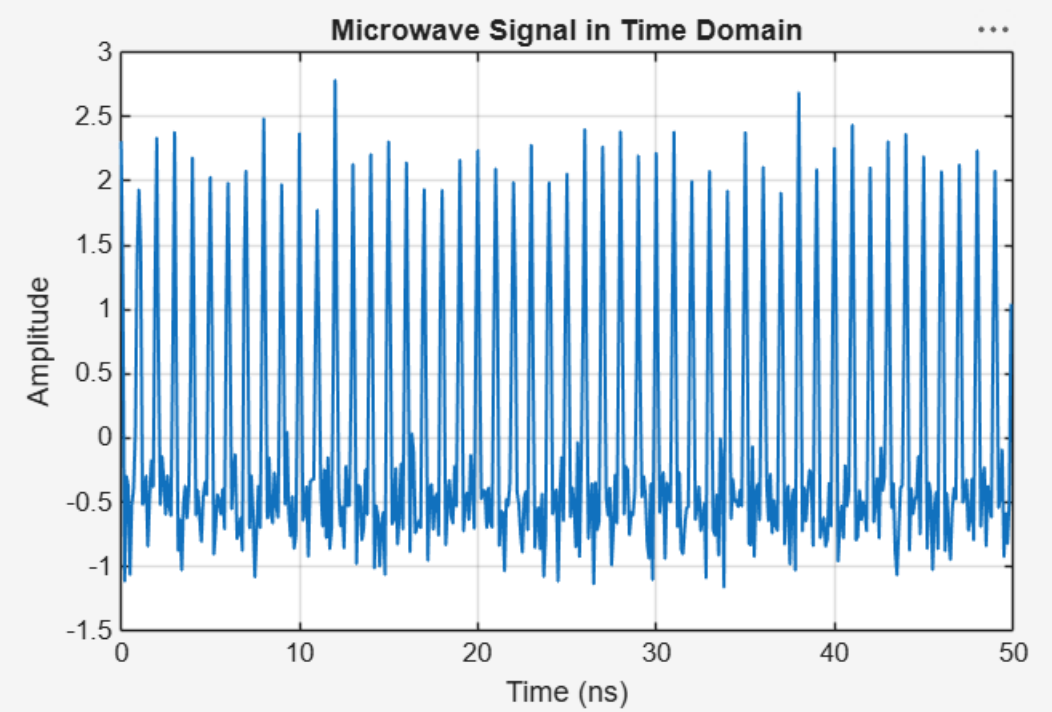
```
x = cos(2*pi*f1*t) + 0.7*cos(2*pi*f2*t) ...
    + 0.5*cos(2*pi*f3*t);
% Add noise
x = x + 0.2*randn(size(x));
% Time-domain signal
figure;
plot(t(1:500)*1e9, x(1:500), 'LineWidth', 1.2);
grid on;
xlabel('Time (ns)');
ylabel('Amplitude');
title('Microwave Signal in Time Domain');
% Frequency-domain analysis
N = length(x);
X = fft(x);
P = abs(X/N);
P = P(1:floor(N/2)+1);
P(2:end-1) = 2*P(2:end-1);
f = Fs*(0:floor(N/2))/N;

% Spectrum analyzer display
figure;
plot(f/1e9, 20*log10(P+1e-12), 'LineWidth', 1.2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Microwave Signal Spectrum');
xlim([0 4]);
```

```
% Find dominant frequency components
[pks, locs] = findpeaks(P, f, ...
    'MinPeakHeight', 0.1, ...
    'MinPeakDistance', 0.3e9);
fprintf('Dominant Frequency Components:\n');
for k = 1:length(locs)
    fprintf('Frequency = %.2f GHz, Amplitude = %.3f\n', ...
        locs(k)/1e9, pks(k));
end
```

Output:





Result:

The microwave signal is simulated, and its frequency components are identified using FFT-based spectral analysis in MATLAB.